



MULTIMEDIA UNIVERSITY OF KENYA

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

UNIVERSITY EXAMINATIONS 2023/2024

SECOND YEAR SUPPLEMENTARY/ SPECIAL EXAMINATION FOR THE DEGREE
OF BACHELOR OF SCIENCE IN COMPUTER SCIENCE/ BACHELOR OF SCIENCE
IN SOFTWARE ENGINEERING/ BACHELOR OF SCIENCE IN INFORMATION
TECHNOLOGY/ BACHELOR OF SCIENCE IN COMPUTER TECHNOLOGY

ICS 2205/BIT 2224/CCS 2215/BCT 2206:

DIGITAL LOGIC / DIGITAL ELECTRONICS

DATE: JULY 2024

TIME: 2 HOURS

INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED.

ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS.

QUESTION ONE (30 MARKS)

- a) Highlight the characteristics of the octal number system. (4 marks)
- b) Perform the following conversions. (9 marks)
- i) 25.625₁₀ to binary
 - ii) 103₁₀ to Hexadecimal
 - iii) 1FB₁₆ to octal
- c) For the logic circuit shown below, obtain the expression for the output X and draw a truth

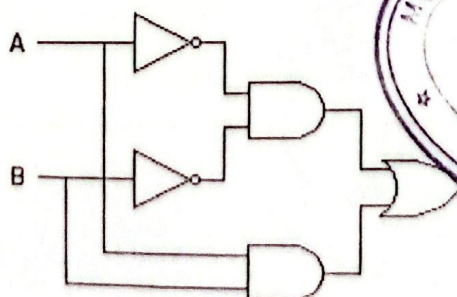


table for the gate.

(8 marks)

d) Perform the following operations.

(9 marks)

i) $100F_{16} - 275_{16}$

ii) $1741_8 - 1001100111_2$

iii) $OCF_{16} + 235_{10}$

QUESTION TWO (TWENTY MARKS)

a) Differentiate between asynchronous and synchronous sequential logic circuits. (4 marks)

b) Perform the following operations:

i) 10010_2 AND $(01101_2$ OR $11001_2)$

(5 marks)

ii) 1101_2 XOR 01101_2 .

(5 marks)

iii) $010110_2 - 110010_2$ using the 2's complement

(6 marks)

QUESTION THREE (TWENTY MARKS)

a) Simplify the expressions using Boolean algebra.

(8 marks)

i) $ABC\{BD + CDE\} + AC$

ii) $ABC + BC + ABC + ABC$

b) Draw a logic circuit which will generate the function $F = B(A + C) + A.B$ using only NAND gates.

(6 marks)

c) From the truth table below, determine the expression for the output x and draw the implementation of a simplified expression of X

(6 marks)

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0



0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

QUESTION FOUR (TWENTY MARKS)

- With the aid of a logic diagram explain the operation of a full adder. (6 marks)
- Implement the expression $y = ABC + AB + BC$ using an AND gate and an OR gate. (4 marks)
- Explain the operation of a falling edge triggered flip flop. (5 marks)
- Design a combinational logic circuit for binary to gray code conversion. (5 marks)

QUESTION FIVE (TWENTY MARKS)

- Minimize the following expressions using k-maps (10 marks)
 - $D = ABC + ABC + ABC + ABC + ABC + ABC + ABC + ABC$
 - $X = AB\{B + A(B + C)\} + BC$
- Prove the following laws of Boolean Algebra (8 marks)
 - $A + AB = A + B$
 - $A + BC = (A + B)(A + C)$
- Highlight the characteristics of the excess 3 number systems. (2 marks)





MULTIMEDIA UNIVERSITY OF KENYA
FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY
UNIVERSITY EXAMINATION 2020/2021
SECOND YEAR XXXX SEMESTER SPECIAL / SUPPLEMENTARY EXAMINATIONS
FOR THE DEGREE OF BACHELOR OF SCIENCE IN COMPUTER TECHNOLOGY

BCT 2206: DIGITAL ELECTRONICS

DATE: 15TH JUNE, 2022

TIME: 2 HOURS

INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED
ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS

QUESTION ONE (THIRTY MARKS)

- a. Define the term *Radix* in number systems. (1 mark)
- b. Give one difference between Latch and a flip flop (1 mark)
- c. Give one advantage of binary number system (1 mark)
- d. Show how the NAND universal gates can be used to realize the AND logic operation (1 mark)
- e. Express $6A32.5F_{16}$ as the sum of product of its weighted basis (2 marks)
- f. Using Boolean algebra techniques simplify this expression and implement its logic circuit
$$A.B + A(B + C) + B(B + C)$$
 (2 marks)
- g. Find the gray code of binary number 110110100011 (1marks)
- h. Find the excess-3 equivalent of $(237.75)_{10}$ (3 marks)
- i. Convert the decimal number 9673 to BCD. (2 marks)
- j. State the DE Morgan's theorems (2 marks)
- k. Determine the hexadecimal equivalent of $0100\ 0011\ 0110\ 0010_{BCD}$ to hexadecimal (1 marks)

- l. Convert (+15) and (-22) signed decimal numbers into two's complement and hence determine the solution of 15-22 (3 marks)
- m. Give one differentiate between synchronous and asynchronous counters (1 mark)
- n. Show the logic arrangement for implementing a four-input AND gate using two-input AND gates only. (2 marks)
- o. Consider an arbitrary number system with the independent digits as 0, 1 and X. What is the radix of this number system? List the first 9 numbers in this number system. (2 marks)
- p. Implement the expression $X = \overline{A}\overline{B}\overline{C} + (D + E)$ using logic gates (2 marks)
- q. Sketch the R-2R digital to analogue converter hence determine the analogue equivalent of $(1111)_2$ given that the $V_{cc} = 5\text{ V}$. (3 marks)

QUESTION TWO (TWENTY MARKS)

- a. By use of a logic diagrams and a truth table describe the four types of flip flops and show how the JK flip flop makes the indeterminate state of SR flip flop useful. (8 marks)
- b. Using combined logic circuits with a truth table implement a circuit that can be used as a full adder (6 marks)
- c. Suppose the cash room at a store has access restricted to certain employees, each of who has a key, which produces a logic 1 at particular inputs to an unlocking circuit. Only the store manager (M) can enter alone. The assistant manager (A) and the cashier (C) also have access but only when accompanied by each other, or by the store manager. Write a truth table and the Boolean expression then design a minimized combinational logic circuit using AND and OR gate only that will allow access by producing a logic 1 when the above conditions are met. (6 marks)

QUESTION THREE (TWENTY MARKS)

- a. By use of state and block diagrams, briefly describe the two types of Finite state machines (FSM) (6 marks)
- b. Simplify the following Boolean expressions: (6 marks)
 - i. $A.B.C + A.B.\overline{C} + A.\overline{B}.C + A.\overline{B}.\overline{C} + \overline{A}.B.C + \overline{A}.B.\overline{C} + \overline{A}.\overline{B}.\overline{C} + \overline{A}.\overline{B}.C$

- ii. $(\bar{A} + B + C).(\bar{A} + B + \bar{C}).(C + D).(C + D + E)$
- c. Figures 1 (a) and (b) are circuit diagrams of switching systems. For each circuit, determine the Truth table and the Logic expression hence implement them using NAND gates only (8 marks)

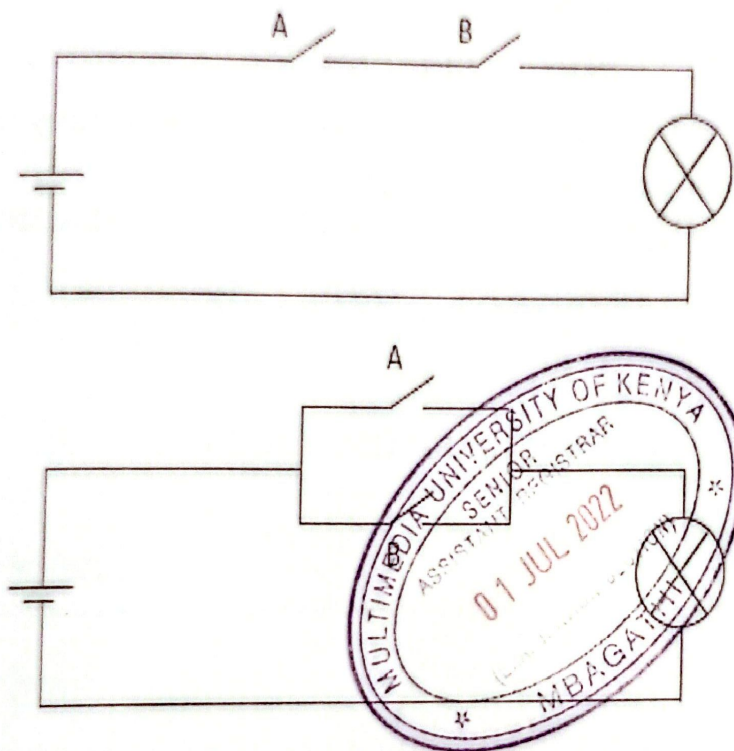


Fig: 1

QUESTION FOUR (TWENTY MARKS)

- a. Implement a 2 to 4 decoder using two NOT gates and four AND gates, hence write Boolean expression and truth table (6 marks)
- b. Define a register and describe any three types of shift registered in sequential circuits (7 marks)
- c. Minimize the following functions using K-map simplification techniques. (7 marks)
- $\bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}\bar{C}$
 - $f(w,x,y,z) = (0,1,2,4,5,6,8,9,12,13,14)$

QUESTION FIVE (TWENTY MARKS)

a. (i) Give one difference between register and counter circuits.

(1 mark)

(ii) Design an asynchronous 3 bit up counter and draw its timing diagram with a truth table for each clock to show how it counts digits 0 to 7.

(6 marks)

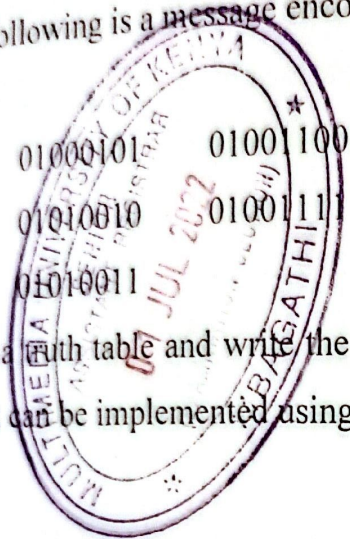
b. The following is a message encoded using ASCII code. Decode the message

(5 marks)

01000101	01001100	01000101	01000011	01010100
01010010	01001111	01001110	01001001	01000011
01010011				

c. Draw a truth table and write the Boolean function of an octal to binary encoder hence show how it can be implemented using OR gates only

(8 marks)





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FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY
UNIVERSITY EXAMINATIONS 2020/2021
SECOND YEAR SECOND SEMESTER SUPPLEMENTARY EXAMINATION FOR
THE DEGREE OF BACHELOR SCIENCE IN COMPUTER SCIENCE/
BACHELOR SCIENCE IN SOFTWARE ENGINEERING

ICS 2205 / CCS 2215: DIGITAL LOGIC/DIGITAL ELECTRONICS

DATE: 15TH JUNE, 2022

INSTRUCTIONS:

TIME: 2 HOURS

ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED.
ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS

QUESTION ONE (THIRTY MARKS)

a) Complete the following conversions

Decimal	BCD	Binary	Gray code
90			
	011100100010	1011101	110101

[10 marks]

b) Using appropriate truth table and Boolean expressions design and implement a 8:1 multiplexer. [10 marks]

c) Draw the logic circuit realization of the following Boolean expression as stated. [5 marks]

$$Y = f(A, B, C) = \overline{(A + B)}(\overline{B} + C)$$

d) Simplify the following expression

$$f(w, x, y) = w\overline{x}y + wx + w\overline{y} + wx\overline{y} \quad [5 \text{ marks}]$$

QUESTION TWO [TWENTY MARKS]

- Using a truth table and Boolean expressions, design a Full-adder. [7 marks]
- What do you mean by a digital decoder? Draw the logic circuit for a 8-1 decoder using basic gates and briefly explain its operation. [5 marks]
- Design a logic circuit that has three inputs, A, B, and C, and whose output will be HIGH only when a majority of the inputs are LOW. (Show the truth table and the circuit diagram) [8 marks]

QUESTION THREE [TWENTY MARKS]

- Use K-Maps to reduce the following expression [10 marks]
 $F(w,x,y,z) = \sum m(0,6,8,13,14) + \sum d(12,15)$
- Show how a NAND gate can be used to implement an
 - OR gate [4 marks]
 - NAND gate [6 marks]
- Write the Boolean expression equivalent to the following logic circuit. [6 marks]



QUESTION FOUR [TWENTY MARKS]

- Design a combinational circuit with three inputs and one output. The output is logic 1 when the binary value of the input is less than 3. The output is logic 0 otherwise. [6 marks]
- Design a modulo 16 ripple down counter using T-flip flops [8 marks]
- What is an encoder? Draw the schematic of a general encoder with X inputs. Explain briefly its operation. Give the logic circuit and truth table for an decimal-to-binary simple encoder with active-low inputs. [6 marks]

QUESTION FIVE [TWENTY MARKS]

- Simplify the Boolean expression below [5 marks]

$$(A \cdot B \cdot C)(A + C)(A + \overline{C})$$
- Draw a logic circuit that directly implements the logic function
 $F = (AB' + CD)'E + BC(A + B)$. [5 marks]
- Simplify the expression. [5 marks]
- Implement the simplified expression using NAND gates only [5 marks]



MULTIMEDIA UNIVERSITY OF KENYA
FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY
UNIVERSITY EXAMINATIONS 2018/2019
SECOND YEAR SECOND SEMESTER EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE IN COMPUTER SCIENCE/SOFTWARE
ENGINEERING/COMPUTER TECHNOLOGY

CCS 2215/BCT 2206: DIGITAL ELECTRONICS

DATE: 17TH DECEMBER, 2019

TIME: 2 HOURS

INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN THE ANSWER BOOKLET PROVIDED.
ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS

QUESTION ONE (THIRTY MARKS)

- a) Perform the following binary subtraction using 2's complement arithmetic: $11011 - 11001$.
[3 marks]
- b) Using appropriate truth table and Boolean expressions design and implement a 1:4 Demultiplexer.
[7 marks]
- c) Construct the logic circuits that can implement the Boolean expressions:
i. $Y = \overline{\overline{A}\overline{B}} + AB$ [3 marks]
ii. $Y = (A + B + \overline{C}\overline{D}\overline{E})(\overline{B}\overline{C}\overline{D})$ [5 marks]
- d) Draw a circuit diagram and a truth table of a J-K flip-flop and explain its operation.
[6 marks]
- e) Simplify the following expression
 $F = AB + \overline{A}\overline{C} + A\overline{B}C(AB + C)$ [6 marks]

QUESTION TWO (TWENTY MARKS)

a) Using a truth table and Boolean expressions, design a half-subtractor.

b) What do you mean by a data selector (Multiplexer)? Draw the logic circuit for a four input multiplexer using basic gates and briefly explain its operation. [7 marks]

c) Simplify the following SOP expression using the K-map:

$$Y = B\bar{C}\bar{D} + \bar{A}B\bar{C}D + AB\bar{C}D + \bar{A}BCD + ABCD$$

d) Implement the simplified expression using logic gates [5 marks]

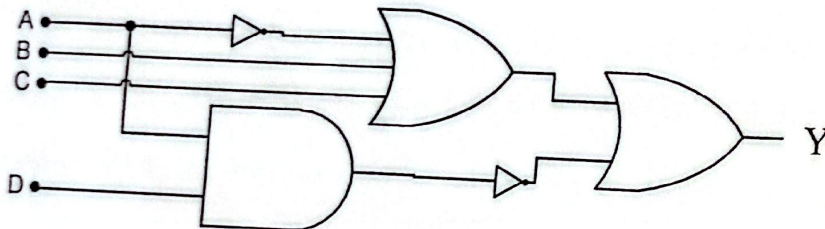
QUESTION THREE (TWENTY MARKS)

a) Draw a four-stage shift-right register using D-flip flops and explain its operation. [5 marks]

b) Using the truth table, show that the Boolean expression $A + BC = (A + B)(A + C)$ [5 marks]

c) What is a full adder? With the aid of a truth table and a suitable diagram, explain how a full adder can be implemented using EX-OR/OR/AND gates. (5marks)

d) Write the Boolean expression for output Y for logic circuit shown in the figure below. [5 marks]



QUESTION FOUR (TWENTY MARKS)

a) Write the binary equivalent of the Gray Code number, 10110101.

b) Given the Boolean expression

$$F = A\bar{B} + AC + \bar{C} + A\bar{B}C + ABC$$

(i) Express in standard SOP form

(ii) Simplify the expression using Boolean laws.

(iii) Using a K-map, simplify the expression

c) What is an encoder? Draw the schematic of a general encoder with X inputs. Explain briefly its operation. Give the logic circuit and truth table for an octal-to-binary simple encoder with active-low inputs. (5marks)

QUESTION FIVE (TWENTY MARKS)

- a) A logic circuit has four inputs A, B, C, D and two outputs Y_1 and Y_2 . Y_1 is high when majority of inputs are high. Y_2 is high only when all inputs are of the same type.
- (i) Draw the truth table of the circuit.
 - (ii) Using a K-map, obtain the expression for each output in Product-of-Sums (POS) form.
- b) Using NAND gates only, implement the simplified logic circuit for both outputs.
- [6 marks]
[6 marks]
[8 marks]



D.K.



MULTIMEDIA UNIVERSITY OF KENYA

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

UNIVERSITY EXAMINATIONS 2020/2021

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SECOND YEAR SECOND SEMESTER EXAMINATIONS FOR THE DEGREE OF
BACHELOR OF SCIENCE IN COMPUTER SCIENCE AND BACHELOR OF SCIENCE
IN SOFTWARE ENGINEERING / BACHELOR OF SCIENCE IN COMPUTER
TECHNOLOGY

BCT 2206/ CCS 2215: DIGITAL ELECTRONICS

0101 100
0101 100

DATE: 7TH DECEMBER 2021

TIME: 2 HOURS

INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED.

ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS.

QUESTION ONE (THIRTY MARKS)

- a) Convert the following to their corresponding bases
- i. $9BCD_{16} = (\dots\dots\dots)_8$ [2 marks]
 - ii. $512_{10} = (\dots\dots\dots)_2$ [2 marks]
 - iii. $323_4 = (\dots\dots\dots)_5$ [3 marks]
- b) Convert 10101111 gray code number to its binary equivalent [5 marks]
- c) Subtract the following using 2's complement
- i. $101_2 - 10110_2$ [4 marks]
 - ii. $54_{10} - 35_{10}$ [4 marks]
- d) Simplify the expression using Boolean algebra and identities
- i. $F(x,y,z) = y(x'z + xz') + x(yz + yz')$ [5 marks]
 - ii. $Y = (A + B)(A + (AB)')C + A'(B + C') + A'B + ABC$ [5 marks]

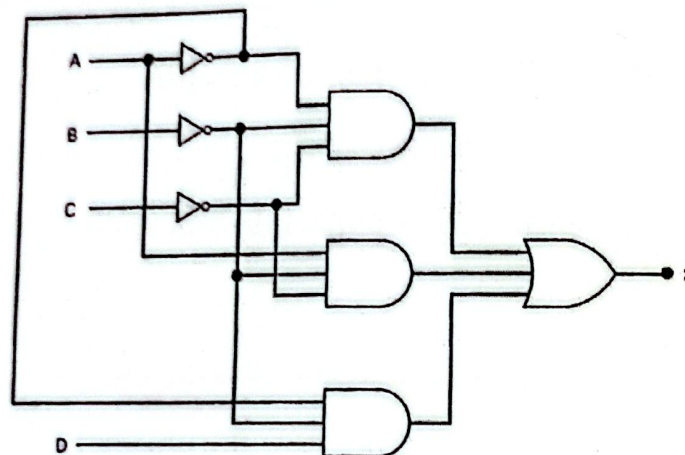
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QUESTION TWO (20 MARKS)

a) Using a truth table and Boolean expressions, design a half-subtractor. [4 marks]

b) Design a code converter that converts a 4-bit number from Gray code to binary code. [10 marks]

c) i) Write the Boolean expression for the circuit shown below. [3 marks]
 ii) Determine the value of x in all possible input conditions and list them in a truth table [3 marks]



QUESTION THREE [20 MARKS]

a) Minimize the following functions using Karnaugh maps

i) $Y(w,x,y,z) = \sum m(2, 4, 6, 7, 8, 9, 10, 11, 12, 14, 15) + \sum d(0,1,3)$ [6 marks]

ii) $F = WX' + YZ + XY' + XZ'$ [6 marks]

b) Implement NOR gate using NAND gates only [4 marks]

c) With the aid of a truth table and a suitable diagram, explain how a full adder can be implemented using EX-OR/OR/AND gates. [4 marks]

QUESTION FOUR [20 MARKS]

a) Design a 16-to-1 line decoder and explain its operation. [6 marks]

b) Design a modulo 16 asynchronous up counter using T-Flip flops [8 marks]

- c) What is a decoder? Draw the schematic of a general encoder with X inputs. Explain briefly its operation. Give the logic circuit and truth table for a 3-to-8 line decoder with active-low inputs. [6 marks]

QUESTION FIVE [20 MARKS]

- a) Minimize the following Boolean expression [6 marks]

$$\overline{(A \cdot B \cdot C)}(A + C)(A + \overline{C})$$

- a) Design a 3 bit magnitude comparator, showing the equations and the circuit diagram for the three possible outputs. [8 marks]

- b) Implement the following function using NAND gates only [6 marks]

$$F = A(B' + C') + BC$$

8	3	9	8	8	5
8	4	0	8	5	5
8	6	2	3		1
8	7	7			7
8	9				5
8	1				1
1	0				1

115715

2	5	1	2	0
2	2	5	6	0
2	1	2	8	0
2	6	4		0
2	3	2		0
2	1	6		0
2	8			0
2	4			0
2	2			0
2	1			1
0				

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